Where Should Domain Safety Policies Live? Policy Placement in LLM Safety Pipelines

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Problem and motivation

A deployed LLM usually sits between an input guard and an output guard, so a benign request can be stopped at any of three stages: the input guard, the model itself, or the output guard. In finance and medicine this boundary is hard: explaining how to spot falsified financial statements should be answered, while helping to falsify them should not.

Deployers can write this boundary as a domain policy, but they must also decide where it goes: the answering model's system prompt, the input guard, the output guard, or a combination. Improving one stage may change little for the user if another stage still blocks the request or lets harm through. Under Topic 4 (Defense Stacking), we study this placement choice and its trade-offs.

Objectives

Our research question: when the same domain policy is given to the answering model, the input guard, or the output guard, which placement delivers more compliant, useful answers while limiting harmful assistance? We aim to:

1. Compare all placement combinations end to end on one financial and one medical education assistant.
2. Attribute each refusal or block to a pipeline stage, and test whether a placement's effect depends on the other stages.
3. Recommend placements for each domain, stating the models, policies and data they apply to.
4. Build an annotated financial evaluation set with allowed and prohibited requests that share topic and wording.

Limitations of previous work

Stacking studies mainly ask which defenses to combine. CASCADE [1] tests 19 attacks and 15 defenses across pipeline stages under a direct jailbreak threat model. Alotaibi et al. [2] measure correlated failures across a seven-layer stack. Their focus is not where one domain policy is placed.

Domain guard work mostly evaluates the detector. Llama Guard [3] allows adjustable categories, and FinGuard [4] adapts to institutional policies, but a better detector verdict need not mean a better delivered answer. Gehrmann et al. [5] report that public guardrails miss most of the financial content risks they study.

Datasets such as Health-ORSC-Bench [6] and FinSafeGuard [7] supply requests for testing models and guards, not placement choices. We do not claim custom-policy guards, paired financial requests or per-layer attribution as new. Our addition is a controlled comparison of placement, using identical policy text at each stage and judging the final answer.

Plan

Design. Each domain has one frozen policy with allow, deny and exception clauses. We test two answering models, Qwen2.5-7B-Instruct and Llama-3.1-8B-Instruct; both guards use Llama Guard 3 8B. For each answering model, three switches each choose default or policy: the model's system prompt (M), the input guard (I) and the output guard (O). This gives eight main configurations, with both guards always present. Two references remove both guards and use the same answering model alone, without and with the policy. The same runs also include a compact

setting for I and O only, with rewritten categories but no full policy text. Counting it, there are 2×3×3 = 18 guarded configurations: the eight main ones plus ten that use a compact guard. The main analysis uses the eight; the ten are a secondary check on prompt form, though they change both length and content. Because guards only pass or block, each request needs two model responses and nine guard decisions per answering model, from which we compose every configuration, including the references.

Data. We will sample boundary prompts from Health-ORSC-Bench [6] and synthetic financial-services requests from FinSafeGuard [7], relabel them against our policies, and add financial requests with similar wording but different intent. Each allowed request records a minimum acceptable answer. The 50 requests already used to revise the policies (25 per domain) stay as development data. The final test set uses 100–200 new requests per domain, keeps their paraphrases and source families out of development data, and reports class balance. Both annotators label test requests independently against the frozen policy and resolve disagreements before evaluation.

Evaluation. The primary outcomes are useful completion on policy-allowed requests and harmful assistance delivered on policy-prohibited requests. Useful completion requires a delivered answer that is compliant, correct and meets the annotated minimum. A fixed MT-bench subset gives only an exploratory signal on general capability; because an education assistant may properly decline out-of-scope questions, we separate scope refusals from lower capability. A frozen rubric, applied by an LLM judge that is neither a tested model nor the guard, will judge refusal, compliance, correctness and completeness separately, with human audit of both compliant and violating judgments. We will report stage attribution and paired comparisons with confidence intervals, resampling related paraphrases together. Deployment cost, which accounts for short-circuiting, is reported separately from offline compute.

Progress. We have a working harness with a self-check, a drafted policy for each domain, and 25 annotated financial requests. Pilot runs found that Llama Guard 3's Hugging Face template silently ignores custom categories, so we build guard prompts ourselves. In an earlier medical pilot on 25 selected requests, the policy prompt unblocked none of the 11 requests provisionally labeled allowed and blocked by the default guard. These labels need re-review, so this is only a hypothesis: placing a policy in the guard prompt may not be enough.

Schedule and risks. Freeze policies and the judge by Nov 10; run development evaluation by Nov 17; freeze main results by Nov 24; present on Dec 1; submit the report on Dec 8. If annotation runs slowly, we will reduce sample size and report wider intervals rather than drop the human audit. A small placement effect would still be a valid result.

References

[1] Luo and Han. CASCADE Against Jailbreaks. arXiv:2609.21793.
[2] Alotaibi et al. Layered LLM Defenses as an Ensemble. arXiv:2608.28327.
[3] Inan et al. Llama Guard. arXiv:2312.06674.
[4] Dou et al. FinGuard. arXiv:2605.29427.
[5] Gehrmann et al. Understanding and Mitigating Risks of Generative AI in Financial Services. arXiv:2504.20086.
[6] Zhang et al. Health-ORSC-Bench. arXiv:2601.17642.
[7] Thacil et al. FinSafeGuard: A Quality-Driven Synthetic Dataset for Financial AI Safety. 2025.